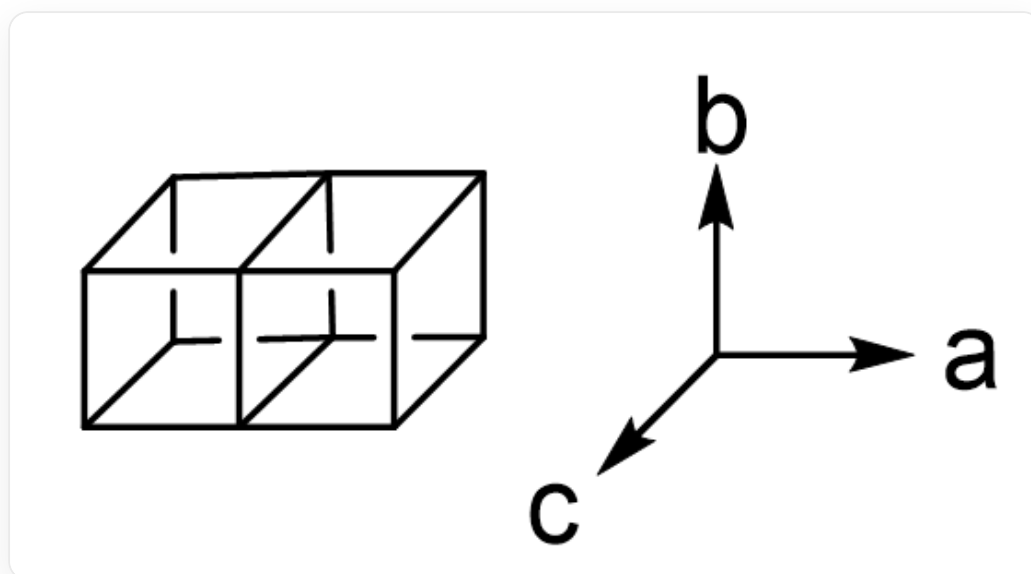


Question

A , **B** and **C** represent an unknown element, respectively. The reaction of A_3C_2 with BC yields a crystal **X** containing **A** , **B** and **C**. **X** belongs to the orthorhombic crystal system with unit cell parameters: $a = 850.3 \text{ pm}$, $b = 1028.4 \text{ pm}$, $c = 503.4 \text{ pm}$, where the arrangement of **A** and **C** is similar to that of NaCl . It can be regarded as two deformed hexahedrons connected coplanarly along the a axis to form the unit shown in the figure below. The two hexahedrons in each unit are then infinitely extended along the c axis in a coplanar manner to form an infinite chain structure; the chains are not directly connected, but form an infinite three-dimensional structure by sharing **C** atoms with the $[\text{BC}_3]$ unit. The $[\text{BC}_3]$ units in **X** are arranged perpendicular to the c axis, the **B** center has a planar triangular configuration with two orientations, and each $[\text{BC}_3]$ unit has a $\text{B} - \text{C}$ bond parallel to the b axis. It is known that all **C** atoms in the crystal participate in the coordination of **B**.



“A schematic diagram. On the left is a structure consisting of two cubes connected coplanarly in the horizontal direction. On the right is a three-dimensional coordinate system with the a -axis pointing to the right, the b -axis pointing upwards, and the c -axis pointing to the front left.”

According to the crystal diffraction results, the coordinates of some **C** atoms in the crystal are: $(0.500, 0.624, 0.750)$, $(0.808, 0.371, 0.750)$; the bond length of the $\text{B} - \text{C}$ bond parallel to the b axis is 186.3 pm . Select the correct $\text{C} - \text{B} - \text{C}$ bond angle from the following options.

A. All other options are incorrect.

B. 113° and 124°

C. 113° and 134°

D. 118° and 124°

E. 118° and 121°

F. 113° and 118°

G. 124° and 134°

H. 121° and 134°

I. 121° and 124°

Answer

Correct Answer: **C**

Detailed Explanation

According to the question, the coordination number of **C** on the shared face of the two parallel hexahedral chains is 6, and the coordinating atoms are 1 **B** and 5 **A**, corresponding to the formation of **B — C** bonds parallel to the *b* axis; the coordination number of **C** on the side faces is 5, and the coordinating atoms are 1 **B** and 4 **A**.

Therefore, each **C** atom in the crystal is connected to 1 **B** atom, and there are 3 **C** atoms around 1 **B** atom, so the ratio of the number of **B** atoms to **C** atoms is 1 : 3. Since **A** and **C** are alternately arranged in the parallelepiped, the ratio of the two is 1 : 1. In summary, the crystal chemical formula is **A₃BC₃**.

CHECKPOINT

1 PTS

There are two coordination forms of **C** atoms, one with a coordination number of 6, and the coordinating atoms are 1 **B** and 5 **A**

CHECKPOINT

1 PTS

The other coordination number of **C** atoms is 5, and the coordinating atoms are 1 **B** and 4 **A**

CHECKPOINT

1 PTS

The crystal chemical formula is $\mathbf{A_3BC_3}$

According to the question, the hexahedral chain in the direction perpendicular to the c axis satisfies translational symmetry, so the lattice type of the crystal belongs to base-centered orthorhombic. Combining the translational symmetry of the space lattice, the coordinates of the $\mathbf{C}$ atom with the same spatial environment as the $\mathbf{C}$ atom at the position (0.808,0.371,0.750) are (0.308,0.871,0.750).

CHECKPOINT

2 PTS

The lattice type of the crystal is base-centered orthorhombic

Assuming the coordinate of a $\mathbf{B}$ atom in the crystal is x , the bond length of the $\mathbf{B} - \mathbf{C}$ bond parallel to the b axis can be obtained as:

$$186.3\text{pm} = (x - 0.624) \times 1028.4\text{pm}$$

Substituting to solve for $x = 0.805$, so the coordinate of the $\mathbf{B}$ atom is (0.500,0.805,0.750).

CHECKPOINT

2 PTS

The coordinate of the $\mathbf{B}$ atom connected to the $\mathbf{C}$ atom with coordinate (0.500,0.624,0.750) is (0.500,0.805,0.750)

Now calculate the bond angle based on the coordinates:

$$\theta = \arctan[(0.871 - 0.805)b / (0.500 - 0.308)a] + 90^\circ = 113^\circ$$

CHECKPOINT

2 PTS

One $\text{C} - \text{B} - \text{C}$ bond angle is 113°

Since **B** is planar triangular coordination, the other $\text{C} - \text{B} - \text{C}$ bond angle is:

$$360^\circ - 113^\circ \times 2 = 134^\circ$$

CHECKPOINT

2 PTS

The other $\text{C} - \text{B} - \text{C}$ bond angle is 134°

Actually, **A** , **B** and **C** represent elements Ca , Cr and N respectively.